



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary

- Contributions to the entropy
 - Translation
 - Includes the $N!$ factor for indistinguishability.
 - Sackur-Tetrode equation for the entropy of an ideal monatomic gas.
 - Rotation: straightforward if $T \gg \theta_{\text{rot}}$.
 - Vibration: often $S_{\text{vib}} \approx 0$ because typically $T < \theta_{\text{vib}}$.
 - Electronic: $S_{\text{elec}} = Nk \ln g_0$ if there are no low-lying excited states.
- Experimental route to the entropy
 - Integrate $C_p(T) / T$ from absolute zero.
 - Add $\Delta S_{\text{pc}} = \Delta H_{\text{pc}} / T_{\text{pc}}$ for any phase changes.
 - Generally excellent agreement between theory and experiment for gases.
 - “Residual entropy” can lead to disagreement.

Nuclear spin statistics

- Previously we introduced, without justification, *the symmetry parameter, σ* , which was needed for the correct calculation of the rotational partition function.
- In this section we will discover that *this factor originates from a consideration of the symmetry of the molecular wavefunction with respect to exchange of nuclei*. You have already come across another example of this kind of symmetry in the *Pauli Principle* which requires the molecular wavefunction to be anti-symmetric with respect to exchange of electrons.

Nuclear spin statistics

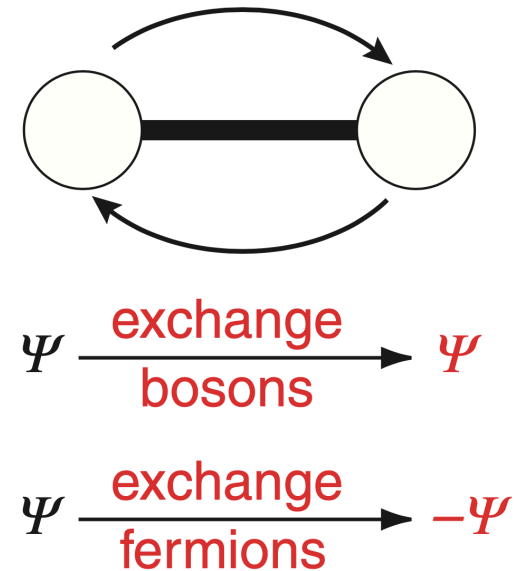
- The starting point for the discussion is *the factorization of the molecular wavefunction into contributions from different kinds of motion*:

$$\Psi = \psi_{\text{electronic}} \psi_{\text{vibrational}} \psi_{\text{rotational}} \psi_{\text{nuclearspin}}$$

- This is *the Born–Oppenheimer separation* which you have encountered before. A wavefunction describing the nuclear spins has been included for this discussion. *Normally we do not need to consider this when thinking about molecular spectra as the influence of the nuclear spins is very small and usually only detectable at the very highest resolution. However, for the present discussion the nuclear spin wavefunction is crucial.*

Nuclear spin statistics

- If we swap two identical nuclei (for example the two ^1H nuclei in $^1\text{H}_2$) then we know that the molecular wavefunction must either remain the same or, at most, change sign. *Remember that a sign change is allowed as ultimately all physical observables depend on Ψ^2 which is unaffected by a sign change in Ψ .* It turns out that nuclei can be classified into two types with respect to exchange: *bosons*, which have the property that the wavefunction remains the same, and *fermions* which have the property that the wavefunction changes sign.
- It is on this apparently simple statement that the whole of this section rests. We first need to look at nuclear spin wavefunctions and their symmetry properties, and then at exactly how this exchange of identical spins is brought about.



Exchanging two identical nuclei either leads to no change in the wavefunction if the nuclei are bosons, or a sign change if the nuclei are fermions.

Nuclear spin wavefunctions

- A nucleus is characterised by *a nuclear spin quantum number, I* , which can have integer or half-integer values. Associated with the spin is *a magnetic moment which can interact with a magnetic field thereby giving rise to $(2I + 1)$ energy levels which are characterised by the quantum number m_I ; m_I takes values between $-I$ and I in integer steps. It is transitions between these energy levels which are observed in NMR spectroscopy. *In the absence of a magnetic field these levels are degenerate; we say that the spin I has associated with it a degeneracy of $(2I + 1)$.**

Nuclear spin wavefunctions

- In a diatomic, the first nucleus has $(2I_1 + 1)$ levels and the second has $(2I_2 + 1)$. *There is only a very weak interaction between the nuclei, so to an excellent approximation we can specify the m_I value for each of the nuclei separately.* Any value of for the first spin can go with any value of m_I for the second, so that *there are a total of $(2I_1 + 1) \times (2I_2 + 1)$ levels for the diatomic.*
- To take a concrete example, consider $^1\text{H}^{19}\text{F}$; both nuclei have $I = 1/2$ and so there are two values of m_I : $+1/2$, $-1/2$; it is usual to denote these as α and β . There are a total of four nuclear spin wavefunctions for the diatomic and these can be written as

$$\alpha(\text{H})\alpha(\text{F}), \quad \alpha(\text{H})\beta(\text{F}), \quad \beta(\text{H})\alpha(\text{F}), \quad \beta(\text{H})\beta(\text{F})$$

where $\alpha(\text{H})$ means that the ^1H nucleus is in state α and so on.

Nuclear spin wavefunctions

- If we consider a homonuclear diatomic the situation is a little more complex as the two nuclei are identical. Let us suppose that we label the two nuclei 1 and 2. *For a wavefunction to be valid it must either remain the same or at most change sign under interchange of the labels 1 and 2.* For it to do any more would be physically unacceptable *as the nuclei are in fact identical and the labels are merely convenient fictions.*
- So, the following wavefunction is acceptable as there is no sign change on swapping the labels

$$\alpha(1)\alpha(2) \xrightarrow{\text{swap 1 and 2}} \alpha(2)\alpha(1) \equiv \alpha(1)\alpha(2)$$

whereas the next wavefunction is unacceptable as swapping the labels leads to a completely different function.

$$\alpha(1)\beta(2) \xrightarrow{\text{swap 1 and 2}} \alpha(2)\beta(1)$$

Nuclear spin wavefunctions

- In fact there are four wavefunctions which satisfy these requirements; three are symmetric with respect to swapping the labels:

$$\begin{aligned} & \alpha(1)\alpha(2) \\ & \left(1/\sqrt{2}\right) [\alpha(1)\beta(2) + \beta(1)\alpha(2)] \\ & \beta(1)\beta(2) \end{aligned}$$

and one is anti-symmetric with respect to the same process:

$$\left(1/\sqrt{2}\right) [\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

- The factors of $1/\sqrt{2}$ are needed for normalization purposes.

Nuclear spin wavefunctions

$$\begin{array}{c}
 \alpha(1)\alpha(2) \\
 (1/\sqrt{2})[\alpha(1)\beta(2) + \beta(1)\alpha(2)] \quad (1/\sqrt{2})[\alpha(1)\beta(2) - \beta(1)\alpha(2)] \\
 \beta(1)\beta(2)
 \end{array}$$

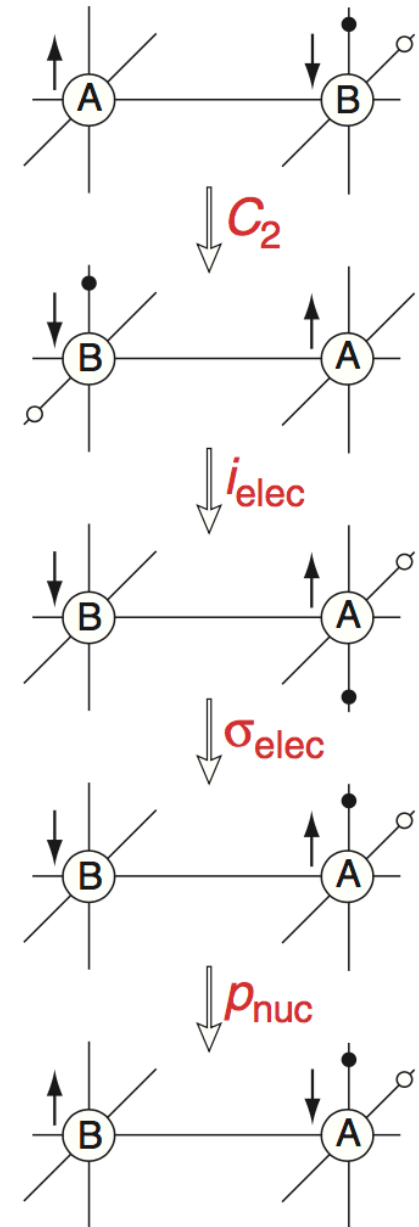
- These wavefunctions above should be familiar as being analogous to those for two electrons, say in the helium atom. In the first case they correspond to the three sublevels of a triplet (**$S = 1$** , **$M_s = 1, 0, -1$**) and in the second case to the one sublevel of a singlet (**$S = 0$** , **$M_s = 0$**).
- *In general, for a homonuclear diatomic, it can be shown that there are $I(2I + 1)$ nuclear spin states which are anti-symmetric with respect to swapping the labels and $(I + 1)(2I + 1)$ symmetric states. The total number of spin states is, as expected, $(2I + 1)^2$.*

How to swap the nuclei

- Recall that *the important thing we need to consider is what happens to the wavefunction when we swap the two nuclei*. However, ‘swapping the nuclei’ is not quite as simple as it sounds, since *what we want to do is swap the nuclei and leave everything else the same*.
- Just rotating the molecule by 180° does swap the nuclei but it also affects other parts of the molecular wavefunction (such as the electronic and rotational part), so we have not achieved what we wanted.
- In fact a more complex series of operations is needed to swap the nuclei and leave everything else the same; these are shown in the figure opposite.

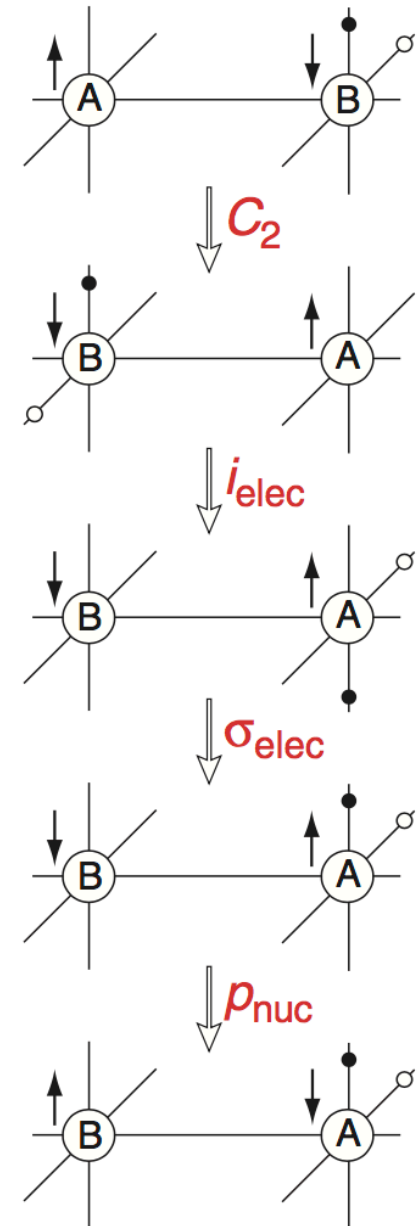
How to swap the nuclei

- The diagram is fairly involved as we have to keep track of everything.*
The letters A and B label the nuclei (they are of course identical, but we need the labels to keep track of them). Also shown by means of the arrows next to the nuclei are the spin states; arbitrarily we have shown A with spin up and B with spin down. Two ‘typical’ electrons are also shown, one by a small filled circle and one by an open circle.



How to swap the nuclei

- The sequence of operations is:
 - Rotate the whole molecule by 180° about an axis perpendicular to the internuclear axis (in Group Theory notation, $\mathbf{C}_2$). This swaps the nuclei, the spin states and moves the electrons.
 - Invert just the electrons through the centre of inversion (i_{elec}).
 - Reflect just the electrons in a mirror plane containing the two atoms (σ_{elec}).
 - Finally, an operation denoted $\mathbf{p}_{\text{nuc}}$ which permutes the nuclear spin states. This means that if spin A is up and spin B is down the result of $\mathbf{p}_{\text{nuc}}$ would be for spin A to be down and B up.



Effect on the molecular wavefunction

- We now have to consider, in turn, the effect of the four operations on the molecular wavefunction.
 - C_2 of the whole molecule
 - Reflection of the electrons through the centre of inversion (i_{elec})
 - Reflection of the electrons through a plane containing the two atoms (σ_{elec})
 - Permutation of the nuclear spin states (p_{nuc})

C_2 of the whole molecule

$$\Psi = \psi_{\text{electronic}} \psi_{\text{vibrational}} \psi_{\text{rotational}} \psi_{\text{nuclearspin}}$$

- *The electronic wavefunction is unaffected by this rotation as the wavefunction depends on ‘where’ the electrons are relative to the nuclei; turning the whole molecule around (nuclei and electrons) does not alter the spatial relationship between the electrons and the nuclei, and so ψ_{elec} is unaffected.*
- *For diatomics the vibrational wavefunction is also unaffected by this rotation of the whole molecule.* Recall that the vibrational wavefunction is expressed in terms of a co-ordinate which is the displacement from the equilibrium position. A positive displacement is still a positive displacement, even if we turn the entire molecule through 180° . *Thus ψ_{vib} is unaffected.*

C_2 of the whole molecule

$$\Psi = \psi_{\text{electronic}} \psi_{\text{vibrational}} \psi_{\text{rotational}} \psi_{\text{nuclear spin}}$$

- *The rotational wavefunctions are affected.* Recall that for the rigid rotor the wavefunctions are the spherical harmonics, which are the same as the angular parts of the hydrogen orbitals. The wavefunction with $\mathbf{J} = \mathbf{0}$ is like an s orbital, and so is invariant to a rotation by 180° . In contrast, if $\mathbf{J} = \mathbf{1}$ the wavefunction is like a p orbital and so rotation by 180° results in a sign change. *In general, for even $\mathbf{J}$ ψ_{rot} is unchanged by C_2 whereas for odd $\mathbf{J}$ there is a sign change.*
- *The nuclear spin wavefunction is unaffected by the rotation*; spin A is still up and spin B is still down.

Reflection of the electrons through the centre of inversion

$$\Psi = \psi_{\text{electronic}} \psi_{\text{vibrational}} \psi_{\text{rotational}} \psi_{\text{nuclearspin}}$$

- *This operation, i_{elec} , clearly only affects ψ_{elec} .* Knowledge of the molecular term symbol will enable us to determine the effect of i_{elec} . If the term symbol has g (gerade) symmetry, then ψ_{elec} is unaffected by i_{elec} ; if the term symbol has u (ungerade) symmetry, then ψ_{elec} changes sign under i_{elec} .

Reflection of the electrons through a plane containing the two atoms

$$\Psi = \psi_{\text{electronic}} \psi_{\text{vibrational}} \psi_{\text{rotational}} \psi_{\text{nuclearspin}}$$

- *This operation, σ_{elec} , clearly only affects ψ_{elec} .*
The superscript + or – from the molecular term symbol will enable us to determine the effect of σ_{elec} . If the superscript is +, then ψ_{elec} is unaffected by σ_{elec} ; if the superscript is – , then ψ_{elec} changes sign under σ_{elec} .

Permutation of the nuclear spin states

- *Under this operation (p_{nuc}), it is clear that symmetric nuclear spin states (as defined in earlier slides) will not change sign whereas anti-symmetric ones will.* For example:

$$\text{symm: } [\alpha(1)\beta(2) + \beta(1)\alpha(2)] \xrightarrow{p_{\text{nuc}}} [\beta(1)\alpha(2) + \alpha(1)\beta(2)]$$

$$\text{anti-symm: } [\alpha(1)\beta(2) - \beta(1)\alpha(2)] \xrightarrow{p_{\text{nuc}}} [\beta(1)\alpha(2) - \alpha(1)\beta(2)]$$

Consequences for $^1\text{H}_2$

- The molecular term symbol for $^1\text{H}_2$ is $^1\Sigma_g^+$; the wavefunction is therefore symmetric with respect to both i_{elec} and σ_{elec} . The nuclei have spin $1/2$, and so there is one anti-symmetric and three symmetric nuclear spin states. We can draw up a table showing the symmetry of the different parts of the wavefunction with respect to the four operations. *There are four possible combinations of odd and even J -states with symmetric and anti-symmetric nuclear spin states.*

rotational	nuclear spin	C_2	i_{elec}	σ_{elec}	p_{nuc}	overall
even J	symmetric	+	+	+	+	+
even J	anti-symmetric	+	+	+	-	-
odd J	symmetric	-	+	+	+	-
odd J	anti-symmetric	-	+	+	-	+

Consequences for $^1\text{H}_2$

- The final column gives the overall result of the four symmetry operations, that is the result of swapping the two nuclei and leaving everything else the same. *The sign in this final column is found by multiplying together the signs from the preceding columns.*
- We finally come to the crucial point: *^1H nuclei are fermions, so swapping them must lead to a change in the sign of the wavefunction.* Therefore the only combinations which are allowed are those in rows two and three.

rotational	nuclear spin	C_2	i_{elec}	σ_{elec}	p_{nuc}	overall
even J	symmetric	+	+	+	+	+
even J	anti-symmetric	+	+	+	-	-
odd J	symmetric	-	+	+	+	-
odd J	anti-symmetric	-	+	+	-	+

Consequences for $^1\text{H}_2$

- We have therefore reached the following remarkable conclusion concerning $^1\text{H}_2$:

Molecules in rotational states with even J -values must be in anti-symmetric nuclear spin states.

Molecules in rotational states with odd J -values must be in symmetric nuclear spin states.

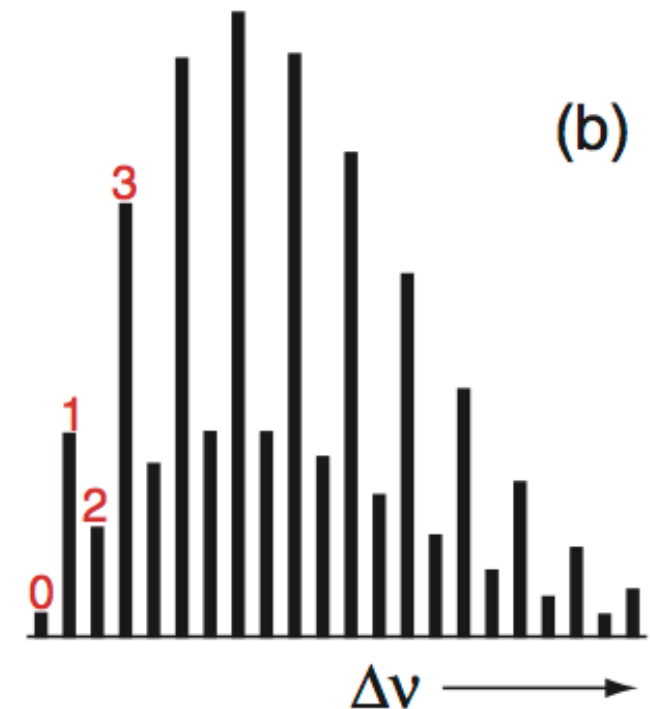
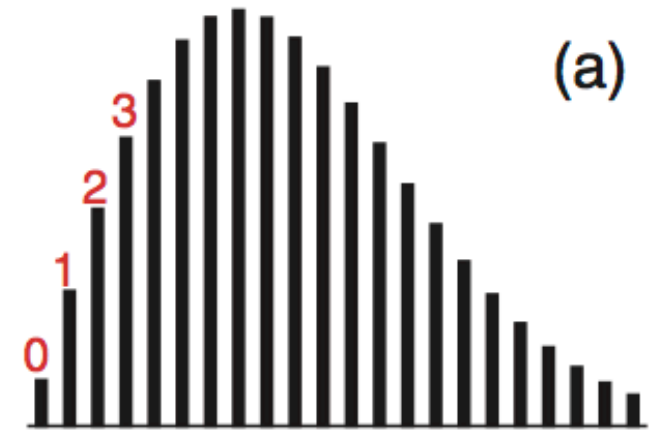
- Recall that there are three symmetric nuclear spin states and one anti-symmetric state, so we can also say that *the molecules with odd J -values have a statistical weight of three when compared to those with even J -values.* This has immediate consequences in the spectra of $^1\text{H}_2$.

Raman spectra of $^1\text{H}_2$

- The rotational Raman spectra of a diatomic shows a series of lines spaced by $4B$, with the first line (the one originating from $J = 0$) being displaced by $6B$ from the laser line. The intensity of these lines depends on the population of the energy levels, and in turn these depend on the degeneracies, $g_J = (2J + 1)$, and the energies, $E_J = BJ(J + 1)$.

$$\text{population} \propto (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right)$$

- Schematic intensity distributions in rotational Raman spectra of $^1\text{H}_2$ (only the set of peaks on one side of the laser line are shown). In (a) the intensities have been calculated without taking into account the effects of nuclear spin.



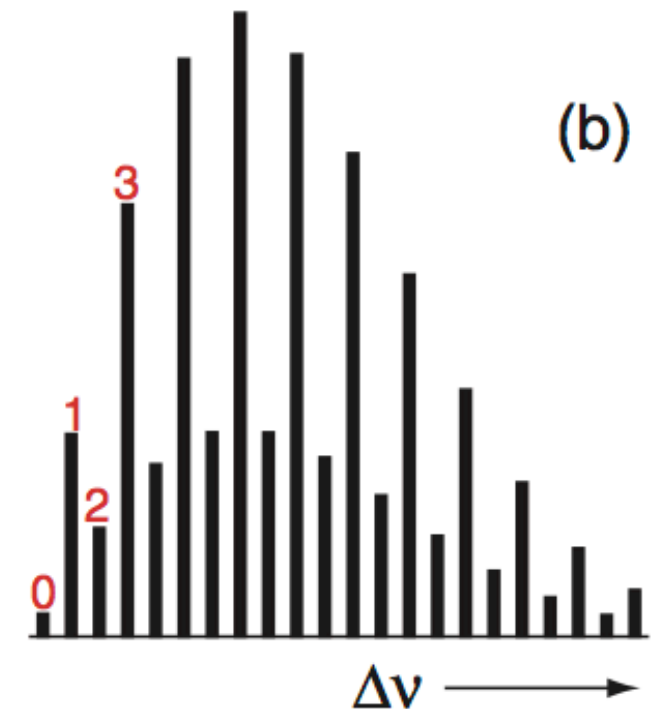
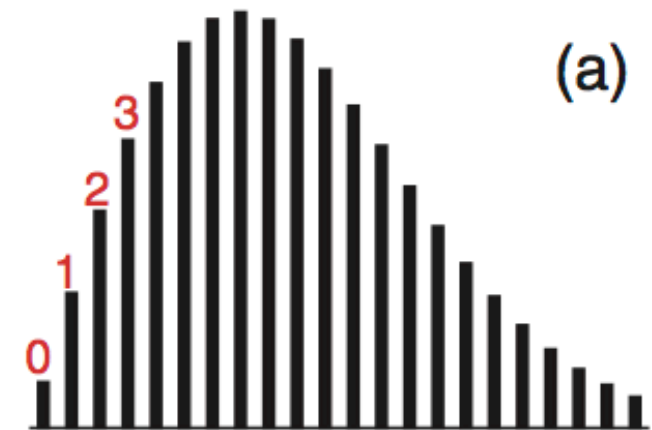
Raman spectra of $^1\text{H}_2$

- We now have an additional factor contributing to the degeneracy: 3 for the odd J -levels and 1 for the even J -levels:

$$\text{odd } J: \quad \text{population} \propto 3 \times (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right)$$

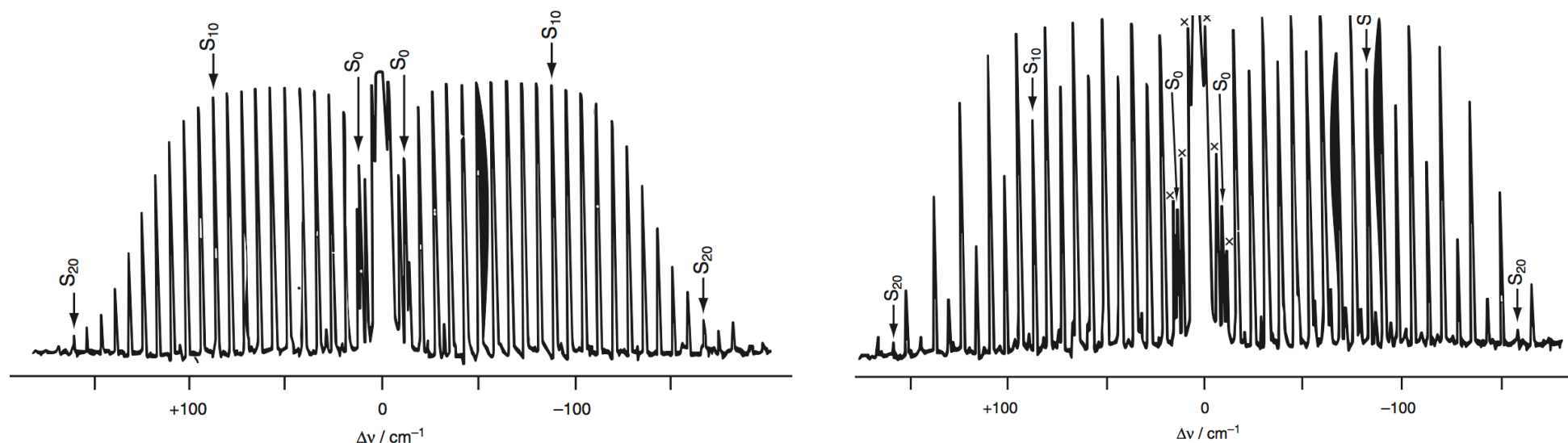
$$\text{even } J: \quad \text{population} \propto 1 \times (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right)$$

- The result is that, setting the other factors aside, lines in the spectra arising from odd J -levels are three times as intense as those from even J -levels. In practice we see an intensity alternation of 1:3:1:3 . . . superimposed on the variation due to the $(2J + 1)$ factor and the exponential term; this is shown in the schematic spectra opposite.
- In (b) these effects of nuclear spin have been included. The numbers are the J values associated with the lines.



Raman spectra of $^1\text{H}_2$

- Other homonuclear diatomics which have a nuclear spin of $1/2$ will show the same intensity alternation: for example $^{15}\text{N}_2$, whose experimental rotational Raman spectrum is compared below with that for $^{15}\text{N}^{14}\text{N}$. *Note that the electronic ground state of N_2 has the same term symbol as that for H_2 , so the argument is identical.*
- Experimental rotational Raman spectra of $^{15}\text{N}^{14}\text{N}$ (on the left) and $^{15}\text{N}_2$ (on the right) showing both the Stokes and anti-Stokes peaks. Due to non-linearity in the detector the peak heights of the strongest lines are less than they ought to be, so the intensity envelope is distorted. The notation S_J indicates a transition arising from rotational level J .



The strange case of *ortho*- and *para*-hydrogen

- Previously we saw that *even J-levels can only go with anti-symmetric nuclear spin states, and odd J-levels with symmetric nuclear spin states*. It turns out that although it is easy for a molecule to change its rotational energy (for example, by collisions) it is very difficult for the nuclear spin wavefunction to change from symmetric to anti-symmetric or vice versa. Essentially the only way to get this to happen is to break apart the H–H bond, and under normal circumstances this just does not happen.
- So, in a sample of ‘normal’ hydrogen there are really two different kinds of hydrogen which do not interconvert:
 - *ortho-hydrogen which has odd J-states and symmetric nuclear spin states;*
 - *para-hydrogen which has even J-states and anti-symmetric nuclear spin states.*

The strange case of *ortho*- and *para*-hydrogen

- *Since different energy levels are available to the two kinds of hydrogen, they have different physical properties.*
- For example, at low temperature their heat capacities will be different. *These differences make it possible to separate ortho- and para-hydrogen*, for example by using gas chromatography. We thus have the remarkable and intriguing possibility of being able to create a sample of H_2 in which the spin states are known.

The strange case of *ortho*- and *para*-hydrogen

- Although the two forms do not interconvert under normal circumstances their interconversion can be promoted by passing the gases or liquid over a catalyst which chemisorbs the hydrogen. At the surface the H–H bonds are broken thus destroying the symmetry. When the atoms recombine they may form either *ortho*- or *para*-hydrogen. *As para-hydrogen involves the lowest energy rotational state ($J = 0$), at low enough temperatures we end up with pure para-hydrogen.* By removing the catalyst and then warming up the liquid we can obtain a sample of pure *para*-hydrogen gas at room temperature.
- A paramagnetic catalyst will also cause this conversion; coupling between the electrons in the catalyst and the nuclei causes singlet to triplet transitions.
- Using the techniques introduced in later lectures, we will be able to calculate that *at high enough temperatures the equilibrium ratio of ortho:para is 3:1; not unexpectedly, this is due to the extra statistical weight of 3 which ortho-hydrogen has.*

Consequences for $^{14}\text{N}_2$

- Like H_2 the electronic term symbol for N_2 is $^1\Sigma_g^+$; the wavefunction is therefore symmetric with respect to both i_{elec} and σ_{elec} . *However, ^{14}N has a nuclear spin of 1 and so is a boson; this means that under exchange of the nuclei the overall wavefunction must remain the same.* For two spins with $I = 1$ there are 3 anti-symmetric and 6 symmetric nuclear spin states. The table is as for H_2 :

rotational	nuclear spin	C_2	i_{elec}	σ_{elec}	p_{nuc}	overall
even J	symmetric	+	+	+	+	+
even J	anti-symmetric	+	+	+	-	-
odd J	symmetric	-	+	+	+	-
odd J	anti-symmetric	-	+	+	-	+

Consequences for $^{14}\text{N}_2$

- Recall that the final column gives the overall result of the four symmetry operations, that is the result of swapping the two nuclei and leaving everything else the same. *As the ^{14}N nuclei are bosons there must be no sign change as a result of this swapping which means that the only combinations which are allowed are those in rows one and four of the table.* In other words:
 - *Molecules in rotational states with even J -values must be in symmetric nuclear spin states;*
 - *Molecules in rotational states with odd J -values must be in anti-symmetric nuclear spin states.*
- *It therefore follows that even J -values have a statistical weight of 6 whereas odd J -values have a weight of 3.* The intensity alternation in the rotational Raman spectrum thus goes 6:3:6:3 (or, as we are just interested in the ratio, 2:1: 2:1). In contrast to the spectrum of $^1\text{H}_2$, in $^{14}\text{N}_2$ the stronger lines are from the even J -values.

The case of spin zero

- Some nuclei, such as ^{16}O and ^{12}C have a nuclear spin of zero – there are thus no nuclear spin states to be concerned with and this leads to some strange consequences. *Spin zero nuclei are always bosons.*

The case of spin zero: $^{16}\text{O}_2$

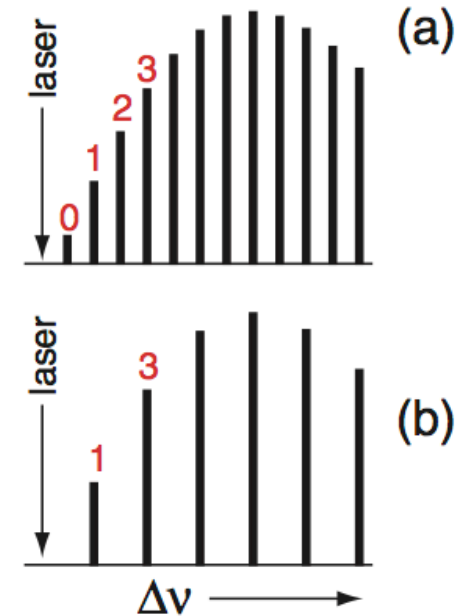
- Consider first $^{16}\text{O}_2$. In contrast to H_2 the term symbol for O_2 is $^3\Sigma_g^-$; the wavefunction is therefore symmetric with respect to i_{elec} but anti-symmetric with respect to σ_{elec} . The table looks somewhat different, therefore:

rotational	nuclear spin	C_2	i_{elec}	σ_{elec}	p_{nuc}	overall
even J		+	+	-		-
odd J		-	+	-		+

- Note that we do not need to consider the operation p_{nuc} as there are no nuclear spin states. Given that ^{16}O is a boson, it is clear that the only allowed combination is that given by the second line of the table. In other words, $^{16}\text{O}_2$ can only exist in odd J -states!

The case of spin zero: $^{16}\text{O}_2$

- This is a truly remarkable result. The application of symmetry principles has led us to the conclusion that $^{16}\text{O}_2$ *can only occupy the odd rotational states*. So, in the rotational Raman spectrum only the lines S_1 , S_3 , S_5 . . . will be seen; the other lines are entirely absent. This means that the separation of the first line from the laser line is not $6B$ but $10B$ and the spacings of the subsequent lines are $8B$, not $4B$.



Schematic rotational Raman spectra of $^{16}\text{O}_2$ (only the set of peaks on one side of the laser line are shown). In (a) the intensities have been calculated without taking into account the effects of nuclear spin. In (b) these effects have been included; only lines originating from odd J -values are seen. The spacing is now therefore $8B$, as opposed to $4B$ in (a).

The case of spin zero: $^{16}\text{O}_2$

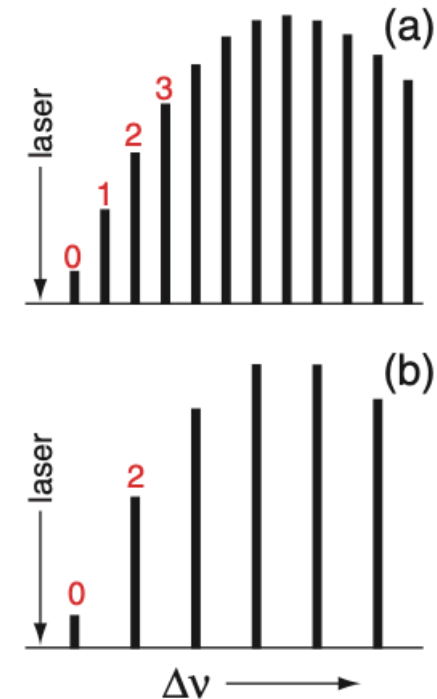
- $^{12}\text{C}^{16}\text{O}_2$ presents a rather different case. The ^{12}C lies at the centre of symmetry, so we can ignore it, so $^{12}\text{C}^{16}\text{O}_2$ can be treated the same way as $^{16}\text{O}_2$. There is a crucial difference, though, which is that the electronic term symbol for CO_2 is $^1\Sigma_g^+$; the wavefunction is symmetric with respect to both i_{elec} and σ_{elec} . The table is:

rotational	nuclear spin	C_2	i_{elec}	σ_{elec}	p_{nuc}	overall
even J		+	+	+		+
odd J		-	+	+		-

- This time the only allowed combination is that given in the first line i.e. $^{12}\text{C}^{16}\text{O}_2$ can only exist in even J -states.

The case of spin zero: $^{12}\text{C}^{16}\text{O}_2$

- So, $^{12}\text{C}^{16}\text{O}_2$ can only occupy the even rotational states. In the rotational Raman spectrum only the lines S_0 , S_2 , S_4 . . . will be seen; the other lines are entirely absent. This means that the separation of the first line from the laser line is $6B$ (as usual) but the spacings of the subsequent lines are $8B$, not $4B$.



Schematic rotational Raman spectra of $^{12}\text{C}^{16}\text{O}_2$ (only the set of peaks on one side of the laser line are shown). In (a) the intensities have been calculated without taking into account the effects of nuclear spin. In (b) these effects have been included; only lines originating from even J -values are seen. The spacing is now therefore $8B$, as opposed to $4B$ in (a).

The case of spin zero: $^{12}\text{C}^{16}\text{O}_2$

- There are some further factors that come into play for CO_2 which distinguish it from a homonuclear diatomic. As we have seen, the asymmetric stretch is IR active and we expect to see rotational fine structure in the form of P and R branches for this parallel vibration. *However, we have just determined that only even J -states are allowed so how can there be any transitions obeying the selection rule $\Delta J = \pm 1$?* If we started in $J = 2$ the selection rule indicates that we would have to move to either $J = 1$ or $J = 3$, but these odd J -states are not allowed.

The case of spin zero: $^{12}\text{C}^{16}\text{O}_2$

- The solution to this apparent paradox is that *for the anti-symmetric stretch although the wavefunction of the ground vibrational state is symmetric, the first excited vibrational state is anti-symmetric*. This introduces another ‘–’ into our table and so the result is that in the $\nu = 1$ state it is the odd J -values which have the correct symmetry. There is therefore no problem in having transitions from even J -values in $\nu = 0$ to odd J -values in $\nu = 1$, as required by the selection rules. The same argument can be extended to the bending mode.
- The symmetric stretch is not IR active, but does give rise to a vibrational Raman spectrum. It turns out that for this mode all of the vibrational states are symmetric so that only even J -states are present. The selection rule is $\Delta J = \pm 2$, so even to even transitions are allowed.

Evaluation of the rotational partition function

- We are now in a position to understand the origin of the symmetry number in the rotational partition function. What the discussion in the previous lectures makes clear is that *for homonuclear diatomics we cannot think about the nuclear spin states and the rotational states separately*. For example, we have seen that in $^1\text{H}_2$ the even J -states go with the anti-symmetric nuclear spin states and that the odd J -states go with the symmetric nuclear spin states.

Evaluation of the rotational partition function

- In earlier lectures, we pointed out that we could find the molecular partition function by multiplying together the partition functions for different sets of energy levels. For example:

$$q = q_{\text{trans}} q_{\text{NS}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

where q_{NS} is the nuclear spin partition function.

- *This separation only works if there are no restrictions on the energy levels which molecules can occupy* i.e. any nuclear spin state can go with any rotational state. Under these circumstances the sums needed to find each of the separate contributions to q can be computed independently of one another.

Evaluation of the rotational partition function

- It is the nuclear spin and rotational contributions which are entangled, so let us for the moment just consider the product $q_{\text{NS}} q_{\text{rot}}$. In a heteronuclear diatomic there are no complications, and we can evaluate these two partition functions separately. As the levels are degenerate we must compute q :

$$q = \sum_j g_j \exp\left(\frac{-\epsilon_j}{kT}\right)$$

- If the two nuclei have spins I_1 and I_2 , then, as explained previously, there are $(2I_1 + 1)(2I_2 + 1)$ nuclear spin states, *all of which are degenerate and which we define to have energy zero.*

Evaluation of the rotational partition function

- Therefore, q_{NS} is simply

$$\begin{aligned}q_{\text{NS}} &= (2I_1 + 1)(2I_2 + 1) \\ &= g_{\text{NS}}\end{aligned}$$

where g_{NS} is the number of nuclear spin states, defined by the above equation.

- The rotational partition function is computed as before, so the product $q_{\text{NS}} q_{\text{rot}}$ is simply

$$q_{\text{NS}} q_{\text{rot}} = g_{\text{NS}} \sum_J (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right)$$

- *For a homonuclear diatomic we have to consider the odd and even J -values separately as each goes with a different set of nuclear spin states:*

$$\begin{aligned}q_{\text{NS}} q_{\text{rot}} &= g_{\text{NS,even}} \sum_{\text{even } J} (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right) \\ &\quad + g_{\text{NS,odd}} \sum_{\text{odd } J} (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right)\end{aligned}$$

- In these equations, $g_{\text{NS,even}}$ is the number of nuclear spin states which go with the even J -states, and likewise $g_{\text{NS,odd}}$ the number which go with the odd J -states.

Evaluation of the rotational partition function

- It is easy to convince ourselves that in the limit $kT \gg B$, where many terms contribute to the sums, the sum over just the even or odd J -states is equal to half the sum over all the J -states.

$$\begin{aligned} \sum_{\text{even } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) &= \sum_{\text{odd } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) \\ \sum_{\text{even } J \text{ or odd } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) &= \frac{1}{2} \sum_{\text{all } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) \\ &= \frac{kT}{2B} \end{aligned}$$

where we have used the result for the rotational partition function from $q_{\text{rot}} = \frac{kT}{\sigma B}$.

- So, *in this limit, for a homonuclear diatomic we have*

$$\begin{aligned} \text{homonuclear: } q_{\text{NS}} q_{\text{rot}} &= g_{\text{NS,even}} \frac{kT}{2B} + g_{\text{NS,odd}} \frac{kT}{2B} \\ &= (g_{\text{NS,even}} + g_{\text{NS,odd}}) \frac{kT}{2B} \end{aligned}$$

whereas for a heteronuclear diatomic we have

$$\text{heteronuclear: } q_{\text{NS}} q_{\text{rot}} = g_{\text{NS}} \frac{kT}{B}$$

- The total number of nuclear spin states is the same in each case (assuming that the nuclear spins are the same), so $g_{\text{NS}} = (g_{\text{NS,even}} + g_{\text{NS,odd}})$. It therefore follows that for a homonuclear diatomic the product $q_{\text{NS}} q_{\text{rot}}$ is one half of that for an equivalent heteronuclear diatomic.

Evaluation of the rotational partition function

- For virtually all calculations we leave out the contribution to the partition function from nuclear spins. *When computing the energy or heat capacity q_{NS} does not make a contribution* as the nuclear spin levels all have zero energy. *In principle these levels ought to contribute to the entropy* as even though their energies are zero they are degenerate.
- However, for reasons which are too complex to go into here these contributions are not included in practical calculations. We will see later on that in the calculation of equilibrium constants these q_{NS} will always cancel out.

Evaluation of the rotational partition function

- So, leaving out the nuclear spin part we have:

$$\text{heteronuclear: } q_{\text{rot}} = \frac{kT}{B} \quad \text{homonuclear: } q_{\text{rot}} = \frac{kT}{2B}$$

- *This, then, is the origin of the symmetry parameter, σ , introduced in earlier lectures. In summary, in a homonuclear diatomic there are restrictions on which rotational states can combine with which nuclear spin states. As a result, fewer combinations of levels are available to the molecule and so the partition function is less than in an equivalent heteronuclear diatomic where no such restrictions exist.*
- Similar arguments can be put forward for more complex molecules. For symmetrical linear molecules (such as CO_2 and HCCH) $\sigma = 2$ by analogy with the case of a homonuclear diatomic. For non-linear molecules it turns out that σ is equal to the number of equivalent positions into which the molecule can be rotated, so it is 2 for H_2O and 3 for NH_3 .

Course outline

- Chemical equilibrium

Course outline

- One of the most remarkable applications of statistical thermodynamics is the calculation of equilibrium constants. We will show in this section how it is possible to calculate these just from a knowledge of the energy levels of the reacting molecules. *The results in this section apply only to indistinguishable molecules.*

Chemical potential

- As we saw in this lecture, we need to introduce the concept of *chemical potential*, μ , to deal with the thermodynamics of mixtures. The key point is that the Gibbs energy of a mixture can be computed from

$$G = \sum_i n_i \mu_i$$

where n_i is the number of moles of substance i whose chemical potential is μ_i .

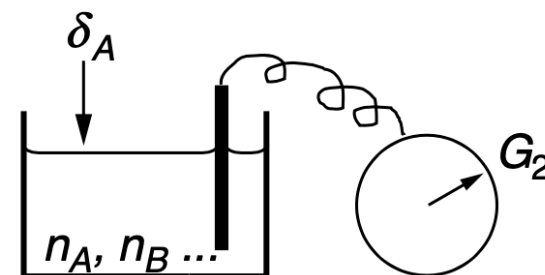
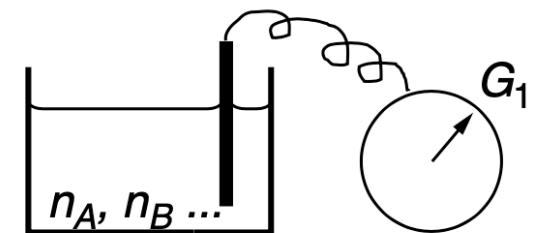
- This relationship shows us that *the Gibbs energy of a mixture depends on its composition* i.e. how many moles of each substance there are present. *In addition, the chemical potentials themselves also depend on the composition of the mixture.*

Chemical potential

- The chemical potential was defined before for substance i as

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{p, T, n_j}$$

- In words, *the chemical potential of i is the change in G when an infinitesimal number of moles of i , dn , is added to the mixture such that the composition remains fixed, all at fixed temperature and pressure.*



$$\mu_A = \frac{G_2 - G_1}{\delta A}$$

Visualization of the meaning of chemical potential. A small amount of substance A is added to a mixture and the increase in G (indicated here on a 'G meter' is recorded. The composition of the mixture remains constant as only an infinitesimal quantity is added.

Chemical potential

- With the definition of chemical potential, we can work out how **G** changes when the amounts of each component are varied:

$$dG = \sum_i \mu_i dn_i \quad \text{const. } T \text{ and } p$$

- In this lecture we used this to find *the relationship between the equilibrium constant, **K**, and the standard Gibbs energy change of a reaction, $\Delta_r G^\circ$:* **$\Delta_r G^\circ = -RT \ln K$** . It turns out there are a number of equivalent definitions of the chemical potential, all of which result in the same properties.

Chemical potential

- The one which we will find most useful is

$$\mu_i = \left(\frac{\partial A}{\partial N_i} \right)_{V,T,N_j}$$

where N_i is the number of molecules (or atoms) rather than the number of moles; note the conditions of constant temperature and volume.

- For a single substance, A is given by (for indistinguishable particles)

$$A = -NkT \ln q + kT (N \ln N - N)$$

- Hence, the chemical potential is

$$\begin{aligned} \mu &= \left(\frac{\partial A}{\partial N} \right)_{T,V} = \frac{\partial}{\partial N} (-NkT \ln q + kT (N \ln N - N)) \\ &= -kT \ln q + kT \ln N + \frac{NkT}{N} - kT \\ &= -kT \ln \frac{q}{N}. \end{aligned}$$

Chemical potential

- In the case that we have a mixture of substances which are non-interacting, we expect a similar expression for the chemical potential of each substance. The argument is that, as there are no interactions, each substance contributes to the total **A** in a way which is independent of the others. So, we write

$$\mu_i = -kT \ln \frac{q_i}{N_i}$$

where **q_i** is the partition function of molecule (or atom) *i* and **N_i** is the number of such molecules present.

Summary of last lecture

- Symmetry of wavefunctions
 - Nuclei have spin.
Interchange of identical bosons leaves Ψ unchanged.
Interchange of identical fermions changes the sign of Ψ .
- Overall wavefunction must have the right symmetry.
- Swapping nuclear position and spin coordinates:
 - C_2 : rotate entire molecule (odd J states change sign)
 - i_{elec} : invert electrons (ungerade terms change sign)
 - σ_{elec} : reflect electrons (Σ^- terms change sign)
 - p_{nuc} : swap nuclear spins (anti-symmetric spin functions change sign)